In [39]:
```
# Q1. Exploring Data
# Display the first 10 rows of the dataset.
student = {
    "name" : "Saad Ali",
    "age" : 25,
    "CGPA" : 3.79,
    "department": "Software Engineering"
}

key = input("Enter a key: ")

if key in student:
    print(student[key])
else:
    print("Key Not Found")
```

```
Saad Ali
```

In [38]:
```
# Q2. Dataset Information
# Display the number of rows and columns in the dataset.
emp = {
    "name": "Ali",
    "position": "HR",
    "salary": 50000,
    "department": "IT"
}

key = input("Enter a key: ")

if key in emp:
    print(emp[key])
else:
    print("Invalid Key")
```

```
Ali
```

In [ ]:
```python
# Q3. Column Selection
# Display only the student_id, age, and exam_score columns.
contacts = {
    "Sara": "03001234567",
    "Ali": "03111234567",
    "Ahmad": "03211234567"
}

name = input("Enter contact name: ")

if name in contacts:
    print(contacts[name])
else:
    print("Contact Not Found")
```

```
03001234567
```

In [ ]:
```python
# Question 4
students = {
    "Ali": {
        "age": 21,
        "cgpa": 3.7,
        "department": "CS"
    },
    "Ahmad": {
        "age": 22,
        "cgpa": 3.4,
        "department": "AI"
    },
    "Sara": {
        "age": 20,
        "cgpa": 3.9,
        "department": "SE"
    }
}

name = input("Enter student name: ")

if name in students:
    print(students[name])
else:
    print("Student Not Found")
```

```
{'age': 20, 'cgpa': 3.9, 'department': 'SE'}
```

In [ ]:
```python
# Question 5
student = {
    "name": "Ali",
    "age": 23,
    "department": "CS"
}

# 1. Display all keys
print(student.keys())

# 2. Display all values
print(student.values())

# 3. Display all key-value pairs
print(student.items())

# 4. Add cgpa using update()
student.update({"cgpa": 3.5})

print(student)
```

```
dict_keys(['name', 'age', 'department'])
dict_values(['Ali', 23, 'CS'])
dict_items([('name', 'Ali'), ('age', 23), ('department', 'CS')])
{'name': 'Ali', 'age': 23, 'department': 'CS', 'cgpa': 3.5}
```

In [10]:
```python
# Question 6
products = {
    "Laptop": {
        "price": 120000,
        "quantity": 5
    },
    "Mouse": {
        "price": 1500,
        "quantity": 10
    },
    "Keyboard": {
        "price": 3000,
        "quantity": 0
    }
}

name = input("Enter product name: ")

if name in products:
    if products[name]["quantity"] > 0:
        print("Product Added to Cart")
    else:
        print("Out of Stock")
else:
    print("Product Not Found")
```

```
Product Added to Cart
```

In [13]:
```
# Pandas Question 1
import pandas as pd

df = pd.read_csv("Student_data_clean.csv")
df.head(10)
```

Out[13]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
0	S1000	21.048994	Female	0	0.029056	4.799806	No
1	S1001	20.496321	Female	ten	3.539930	3.400655	No
2	S1002	20.496321	Male	ten	10.270434	1.780307	No
3	S1003	25.350557	Female	ten	0.816828	5.861446	No
4	S1004	17.937754	Female	ten	5.320886	0.000000	No
5	S1005	23.193687	Male	7.2	0.762742	4.156124	No
6	S1006	22.062970	Female	5.6	0.000000	0.000000	No
7	S1007	20.496321	unknown	4.3	0.000000	3.373847	No
8	S1008	20.496321	Female	ten	3.663144	6.903434	No
9	S1009	23.669075	Female	4.8	5.199887	0.000000	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
0	86.153500	7.294123	Fair	6.517918	Master	Good	11.789345
1	83.959410	1.711612	Good	4.408856	High School	Average	5.432582
2	92.813065	0.245307	Poor	0.000000	unknown	Poor	-3.244017
3	70.689033	4.632657	Poor	0.581851	Master	Good	5.000688
4	83.959410	5.466443	Fair	5.148622	Master	Good	2.408725
5	82.212347	NaN	Fair	0.000000	Master	Average	4.610438
6	83.959410	0.000000	Good	0.000000	Master	Poor	2.176012
7	76.041926	3.109454	Fair	3.046520	Bachelor	Average	4.307425
8	99.326218	4.400000	Good	4.314662	unknown	Good	5.390964
9	96.742474	7.054295	Good	2.072327	Bachelor	Good	11.481817

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
0	Yes	59.197075	9342
1	No	98.718319	5360
2	No	33.361684	9117
3	Yes	27.830878	2436
4	No	60.686418	7714
5	No	100.000000	1863
6	No	93.537529	4953
7	No	100.000000	2313
8	No	80.297148	5120

	extracurricular_participation	exam_score	random_id
9	Yes	100.000000	2126

In [ ]:
```
# Question 2
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Number of rows: 1153
Number of columns: 17
```

In [15]:
```
# question 3
df[['student_id', 'age', 'exam_score']]
```

Out[15]:

	student_id	age	exam_score
0	S1000	21.048994	59.197075
1	S1001	20.496321	98.718319
2	S1002	20.496321	33.361684
3	S1003	25.350557	27.830878
4	S1004	17.937754	60.686418
...	...	...	...
1148	S1431	20.169749	31.321091
1149	NaN	22.671669	45.230354
1150	S1492	16.382979	70.239270
1151	S1979	20.676107	55.382948
1152	S1902	24.716932	67.190582

1153 rows × 3 columns

In [17]:
```
# Question 4
df.isnull().sum()
```

Out[17]:
```
student_id                     121
age                              0
gender                           0
study_hours_per_day              0
social_media_hours               0
netflix_hours                    0
part_time_job                    0
attendance_percentage            0
sleep_hours                    119
diet_quality                     0
exercise_frequency               0
parental_education_level         0
internet_quality                 0
mental_health_rating             0
extracurricular_participation    0
exam_score                       0
random_id                        0
dtype: int64
```

In [19]:
```
# Question 5
df_cleaned = df.drop_duplicates()

df_cleaned
```

Out[19]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
0	S1000	21.048994	Female	0	0.029056	4.799806	No
1	S1001	20.496321	Female	ten	3.539930	3.400655	No
2	S1002	20.496321	Male	ten	10.270434	1.780307	No
3	S1003	25.350557	Female	ten	0.816828	5.861446	No
4	S1004	17.937754	Female	ten	5.320886	0.000000	No
...	...	...	...	...	...	...	...
1148	S1431	20.169749	Female	0	0.147439	2.220364	No
1149	NaN	22.671669	Female	ten	1.463998	1.937920	No
1150	S1492	16.382979	Male	3.8	0.000000	0.000000	No
1151	S1979	20.676107	Male	1.3	1.741795	0.000000	No
1152	S1902	24.716932	Male	4.1	1.071961	4.197893	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
0	86.153500	7.294123	Fair	6.517918	Master	Good	11.789345
1	83.959410	1.711612	Good	4.408856	High School	Average	5.432582
2	92.813065	0.245307	Poor	0.000000	unknown	Poor	-3.244017
3	70.689033	4.632657	Poor	0.581851	Master	Good	5.000688
4	83.959410	5.466443	Fair	5.148622	Master	Good	2.408725
...	...	...	...	...	...	...	...
1148	82.281313	2.247229	Poor	4.636831	unknown	Average	2.064658
1149	98.556989	5.185472	Fair	3.046520	Master	Good	-0.420232
1150	79.850885	12.592937	Good	0.875599	unknown	Good	4.719068
1151	76.300402	5.437463	Good	6.343053	High School	Poor	4.319858
1152	71.468550	8.437898	unknown	0.827871	Master	Good	5.390964

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
0	Yes	59.197075	9342
1	No	98.718319	5360
2	No	33.361684	9117
3	Yes	27.830878	2436
4	No	60.686418	7714
...	...	...	...
1148	No	31.321091	8669

	extracurricular_participation	exam_score	random_id
1149	No	45.230354	5427
1150	No	70.239270	8290
1151	No	55.382948	2093
1152	No	67.190582	8403

1150 rows × 17 columns

In [25]:
```
# Question no 6
df['sleep_hours'] = df['sleep_hours'].fillna(
    df['sleep_hours'].median()
)
df.isnull().sum()
```

Out[25]:
```
student_id                     121
age                              0
gender                           0
study_hours_per_day              0
social_media_hours               0
netflix_hours                    0
part_time_job                    0
attendance_percentage            0
sleep_hours                      0
diet_quality                     0
exercise_frequency               0
parental_education_level         0
internet_quality                 0
mental_health_rating             0
extracurricular_participation    0
exam_score                       0
random_id                        0
dtype: int64
```

In [27]:
```
# Question 7
df['study_hours_per_day'] = pd.to_numeric(
    df['study_hours_per_day'],
    errors='coerce'
)
print(df['study_hours_per_day'].dtype)
```

```
float64
```

In [28]:
```
# Question no 8
df[df['exam_score'] > 80]
```

Out[28]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
1	S1001	20.496321	Female	NaN	3.539930	3.400655	No
5	S1005	23.193687	Male	7.2	0.762742	4.156124	No
6	S1006	22.062970	Female	5.6	0.000000	0.000000	No
7	S1007	20.496321	unknown	4.3	0.000000	3.373847	No
8	S1008	20.496321	Female	NaN	3.663144	6.903434	No
...	...	...	...	...	...	...	...
1130	S1404	26.059073	unknown	4.6	0.303310	5.110632	No
1131	S1406	20.380449	Female	3.6	6.380638	1.219426	No
1145	S1530	15.523021	Male	NaN	0.000000	2.879551	No
1146	S1455	19.711286	Male	8.3	6.992849	0.264495	No
1147	S1647	20.430457	Female	NaN	4.165038	1.937920	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
1	83.959410	1.711612	Good	4.408856	High School	Average	5.432582
5	82.212347	6.054516	Fair	0.000000	Master	Average	4.610438
6	83.959410	0.000000	Good	0.000000	Master	Poor	2.176012
7	76.041926	3.109454	Fair	3.046520	Bachelor	Average	4.307425
8	99.326218	4.400000	Good	4.314662	unknown	Good	5.390964
...	...	...	...	...	...	...	...
1130	73.867623	3.754503	Good	3.046520	unknown	Average	8.189717
1131	68.916511	0.000000	Poor	0.508253	Bachelor	Good	12.152488
1145	83.959410	2.354807	Good	2.329910	Master	Good	3.383579
1146	87.034990	7.157521	Fair	6.943018	High School	Average	12.083630
1147	83.729647	11.225482	Fair	3.424837	unknown	Average	-2.224919

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
1	No	98.718319	5360
5	No	100.000000	1863
6	No	93.537529	4953
7	No	100.000000	2313
8	No	80.297148	5120
...	...	...	...
1130	Yes	86.057739	9796

	extracurricular_participation	exam_score	random_id
1131	Yes	84.631331	3543
1145	Yes	81.632289	9867
1146	No	100.000000	2370
1147	No	83.828071	1643

325 rows × 17 columns

In [29]:
```
# Question NO 9
df[
    (df['study_hours_per_day'] > 5) &
    (df['sleep_hours'] ⩾ 7)
]
```

Out[29]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
94	S1094	21.877130	Female	5.3	2.350669	3.237168	No
131	S1131	20.106261	Female	7.2	5.547043	0.000000	No
171	S1171	20.119481	Female	5.3	2.888899	0.511368	No
214	S1214	18.872441	Male	5.7	5.414765	0.532280	No
222	S1222	22.194291	Male	6.7	1.406480	0.000000	No
232	S1232	18.784415	Female	6.5	7.203870	1.937920	No
293	S1293	19.556450	Female	7.4	4.421131	0.000000	No
298	S1298	21.636060	Female	5.4	2.611274	1.937920	No
306	S1306	25.733835	Female	6.0	2.767127	1.696367	No
310	S1310	20.494998	other	5.3	0.000000	5.232609	No
334	S1334	20.880612	Female	5.8	6.435063	1.937920	No
356	NaN	17.486032	Male	5.8	1.764736	0.000000	No
359	NaN	20.496321	Male	6.3	0.000000	0.000000	No
366	S1366	27.348863	Male	6.3	1.823476	1.022849	No
367	S1367	11.433406	Male	5.8	2.611274	1.585406	No
398	S1398	23.301959	Male	5.7	0.000000	0.000000	No
455	S1455	22.270798	Male	8.3	5.932053	6.207093	No
464	S1464	16.754015	Male	6.3	6.329983	5.901129	No
467	S1467	15.598833	Male	5.4	0.000000	0.000000	No
500	S1500	18.955051	Male	5.5	2.168226	1.031929	No
514	S1514	15.250609	Female	5.9	4.547371	3.461870	No
517	S1517	19.427826	Female	5.8	1.814388	0.000000	No
547	S1547	18.172251	Male	6.3	11.044803	1.937920	No
595	S1595	17.976623	Male	5.6	3.463348	0.000000	No
606	S1606	22.397474	Male	6.8	2.439326	0.000000	No
640	S1640	20.155345	Female	6.4	4.651400	0.000000	No
652	S1652	20.496321	Female	5.9	2.611274	3.250752	No
667	S1667	16.663527	Female	5.6	1.716976	3.984732	No
715	S1715	23.733864	unknown	5.6	0.000000	2.008336	No
764	S1764	16.323436	Male	6.6	1.268070	0.413540	No
809	NaN	19.091784	unknown	5.2	0.000000	5.759771	No
874	S1874	16.940867	Male	5.1	2.731454	0.307086	No
877	S1874	16.940867	Male	5.1	2.731454	0.307086	No
932	NaN	20.818138	Female	5.7	0.000000	1.937920	No
994	S1991	19.841801	Male	6.0	0.719352	11.653201	No
1025	S1488	15.447670	Male	5.3	0.000000	2.083799	No

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
1042	S1875	15.575249	Female	7.6	0.000000	1.937920	No
1080	S1569	27.826733	Male	6.3	0.000000	2.891885	No
1146	S1455	19.711286	Male	8.3	6.992849	0.264495	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
94	75.824853	9.288359	Fair	0.000000	High School	Average	9.443824
131	100.663067	7.539760	Fair	7.397326	High School	Good	6.543756
171	87.546927	8.354680	Good	0.789210	unknown	Good	5.857575
214	100.970076	7.449997	Fair	2.702082	unknown	Average	-2.972704
222	82.858236	7.633558	Good	9.188993	Bachelor	Good	4.984252
232	77.002531	8.515406	Good	0.000000	Master	Average	6.866407
293	83.959410	7.149998	Fair	3.681476	Bachelor	Good	8.375653
298	85.266722	7.152474	Poor	0.000000	High School	Average	2.289350
306	87.339546	7.028897	unknown	6.103610	unknown	Good	11.953331
310	77.672871	8.416760	unknown	5.014756	Master	Average	7.768439
334	95.556835	9.228266	Fair	3.046520	unknown	Average	9.189547
356	76.615155	7.531532	Good	5.159403	unknown	Average	12.502627
359	81.870000	7.955252	Good	7.624561	Bachelor	Good	-0.321471
366	95.308895	11.234142	Poor	0.000000	High School	Good	9.994945
367	86.219444	9.448685	Poor	3.046520	Bachelor	Good	5.390964
398	74.452002	7.557407	Good	0.000000	Master	Good	5.390964
455	85.858854	9.965060	unknown	8.566157	High School	Average	7.572885
464	76.857898	9.522732	Fair	3.046520	Master	Average	4.362568
467	84.727145	8.462827	Fair	3.046520	High School	Good	3.919697
500	90.325603	7.290401	Fair	0.000000	Bachelor	Good	6.113239
514	78.771129	7.197358	Fair	0.000000	High School	Good	2.073562
517	74.971451	10.119722	Fair	4.296744	Bachelor	Good	3.010474
547	86.354079	8.374308	unknown	6.093581	unknown	Good	6.292493
595	79.443668	9.465043	Fair	7.553705	Bachelor	Average	6.790715
606	97.708262	11.832753	Good	4.688521	Bachelor	Poor	3.018348
640	83.959410	8.733632	Fair	0.000000	High School	Average	8.704788
652	97.864030	10.382998	unknown	8.703376	Bachelor	Average	12.738769
667	98.163704	9.089593	Fair	3.046520	unknown	Good	5.390964
715	88.480987	10.958032	Good	2.316672	Master	Good	12.131053
764	85.055872	8.166162	Fair	0.000000	Master	Poor	5.390964
809	83.959410	8.023848	Good	0.000000	Bachelor	Good	5.390964

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
874	95.648970	8.048016	unknown	2.259864	Master	Average	9.087426
877	95.648970	8.048016	unknown	2.259864	Master	Average	9.087426
932	81.807488	11.032836	Good	0.000000	Master	Average	11.098324
994	86.258303	7.183812	Good	3.046520	High School	Good	3.966371
1025	85.836998	9.971123	Good	6.150076	Bachelor	Average	5.941997
1042	96.367834	10.638115	Poor	1.226477	High School	Good	-0.399984
1080	83.959410	10.407000	Fair	6.012734	unknown	Good	4.231231
1146	87.034990	7.157521	Fair	6.943018	High School	Average	12.083630

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
94	No	71.034216	5793
131	No	95.250721	5072
171	Yes	86.159768	5121
214	No	100.000000	6662
222	No	100.000000	2386
232	No	100.000000	2842
293	Yes	94.735722	5860
298	Yes	88.597503	4356
306	No	100.000000	5221
310	No	96.479537	2990
334	No	98.193412	8961
356	No	71.034216	4858
359	Yes	100.000000	7323
366	Yes	97.687973	3694
367	No	96.678824	2238
398	Yes	100.000000	1369
455	No	100.000000	8031
464	No	71.034216	9911
467	No	99.476519	3968
500	No	87.906617	5530
514	Yes	90.082222	7946
517	No	71.034216	1525
547	No	97.490541	8332
595	No	97.178567	9727
606	No	97.510611	8132
640	No	98.255052	1927

	extracurricular_participation	exam_score	random_id
652	No	100.000000	1519
667	Yes	97.507287	4833
715	No	87.851730	8413
764	No	100.000000	9118
809	Yes	88.300678	2907
874	No	96.000869	9779
877	No	96.000869	9779
932	Yes	85.434794	1068
994	No	87.139597	8368
1025	Yes	100.000000	7000
1042	No	100.000000	7199
1080	Yes	97.376065	9132
1146	No	100.000000	2370

In [30]:
```
# Question no 10
df[df['internet_quality'].isin(['Good', 'Average'])]
```

Out[30]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
0	S1000	21.048994	Female	0.0	0.029056	4.799806	No
1	S1001	20.496321	Female	NaN	3.539930	3.400655	No
3	S1003	25.350557	Female	NaN	0.816828	5.861446	No
4	S1004	17.937754	Female	NaN	5.320886	0.000000	No
8	S1008	20.496321	Female	NaN	3.663144	6.903434	No
...	...	...	...	...	...	...	...
1147	S1647	20.430457	Female	NaN	4.165038	1.937920	No
1148	S1431	20.169749	Female	0.0	0.147439	2.220364	No
1149	NaN	22.671669	Female	NaN	1.463998	1.937920	No
1150	S1492	16.382979	Male	3.8	0.000000	0.000000	No
1152	S1902	24.716932	Male	4.1	1.071961	4.197893	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
0	86.153500	7.294123	Fair	6.517918	Master	Good	11.789345
1	83.959410	1.711612	Good	4.408856	High School	Average	5.432582
3	70.689033	4.632657	Poor	0.581851	Master	Good	5.000688
4	83.959410	5.466443	Fair	5.148622	Master	Good	2.408725
8	99.326218	4.400000	Good	4.314662	unknown	Good	5.390964
...	...	...	...	...	...	...	...
1147	83.729647	11.225482	Fair	3.424837	unknown	Average	-2.224919
1148	82.281313	2.247229	Poor	4.636831	unknown	Average	2.064658
1149	98.556989	5.185472	Fair	3.046520	Master	Good	-0.420232
1150	79.850885	12.592937	Good	0.875599	unknown	Good	4.719068
1152	71.468550	8.437898	unknown	0.827871	Master	Good	5.390964

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
0	Yes	59.197075	9342
1	No	98.718319	5360
3	Yes	27.830878	2436
4	No	60.686418	7714
8	No	80.297148	5120
...	...	...	...
1147	No	83.828071	1643

	extracurricular_participation	exam_score	random_id
1148	No	31.321091	8669
1149	No	45.230354	5427
1150	No	70.239270	8290
1152	No	67.190582	8403

713 rows × 17 columns

In [31]:
```
# question no 11
df.sort_values(
    by='exam_score',
    ascending=False
).head(10)
```

Out[31]: Columns 1–8 of 18

	student_id	age	gender	study_hours_per_day	social_media_hours	netflix_hours	part_time_job
1126	S1318	15.486285	unknown	2.2000	2.703445	3.121706	No
5	S1005	23.193687	Male	7.2000	0.762742	4.156124	No
620	S1620	14.544716	Female	1.4000	2.485553	5.267700	No
638	S1638	19.219215	Female	NaN	0.000000	2.909774	No
56	S1056	20.496321	Male	2.5000	4.903691	0.000000	No
23	S1023	24.238106	Female	11.0276	2.010763	1.764792	No
950	S1947	23.106798	Female	4.5000	1.747211	8.530823	No
952	NaN	17.368829	Female	4.4000	2.450181	0.000000	No
649	S1649	26.679326	Female	4.8000	0.800206	4.488416	No
652	S1652	20.496321	Female	5.9000	2.611274	3.250752	No

Columns 9–15 of 18 · first column repeated

	attendance_percentage	sleep_hours	diet_quality	exercise_frequency	parental_education_level	internet_quality	mental_health_rating
1126	83.959410	0.000000	Fair	4.770018	High School	Poor	6.519449
5	82.212347	6.054516	Fair	0.000000	Master	Average	4.610438
620	90.446423	6.054516	Fair	2.887417	unknown	Average	5.390964
638	74.302331	5.547819	Fair	3.046520	unknown	Good	11.926801
56	87.612956	4.189053	Good	7.151303	Bachelor	Average	9.384299
23	91.821104	6.054516	Fair	1.900198	Bachelor	Good	-1.773977
950	69.683420	6.468704	Good	0.255055	High School	Good	3.772476
952	98.583189	0.000000	Fair	5.833283	Bachelor	Good	5.390964
649	83.959410	0.621077	Fair	5.127831	Bachelor	Average	5.402677
652	97.864030	10.382998	unknown	8.703376	Bachelor	Average	12.738769

Columns 16–18 of 18 · first column repeated

	extracurricular_participation	exam_score	random_id
1126	No	100.0	8492
5	No	100.0	1863
620	No	100.0	9881
638	Yes	100.0	2592
56	No	100.0	1084
23	No	100.0	8818
950	Yes	100.0	2059
952	No	100.0	7794
649	No	100.0	8275

	extracurricular_participation	exam_score	random_id
652	No	100.0	1519

In [32]:
```
# question 12
df.nlargest(5, 'exam_score')[
    ['student_id', 'study_hours_per_day', 'exam_score']
]
```

Out[32]:

	student_id	study_hours_per_day	exam_score
5	S1005	7.2000	100.0
7	S1007	4.3000	100.0
9	S1009	4.8000	100.0
23	S1023	11.0276	100.0
56	S1056	2.5000	100.0

In [33]:
```
# Question 13
df['total_screen_time'] = (
    df['social_media_hours'] + df['netflix_hours']
)

df[['social_media_hours', 'netflix_hours', 'total_screen_time']]
```

Out[33]:

	social_media_hours	netflix_hours	total_screen_time
0	0.029056	4.799806	4.828862
1	3.539930	3.400655	6.940585
2	10.270434	1.780307	12.050742
3	0.816828	5.861446	6.678273
4	5.320886	0.000000	5.320886
...	...	...	...
1148	0.147439	2.220364	2.367803
1149	1.463998	1.937920	3.401919
1150	0.000000	0.000000	0.000000
1151	1.741795	0.000000	1.741795
1152	1.071961	4.197893	5.269854

1153 rows × 3 columns

In [ ]:

```
# question 14
def assign_grade(score):
    if score ⩾ 85:
        return 'A'
    elif score ⩾ 70:
        return 'B'
    elif score ⩾ 55:
        return 'C'
    elif score ⩾ 40:
        return 'D'
    else:
        return 'F'


df['grade'] = df['exam_score'].apply(assign_grade)

df[['exam_score', 'grade']]
```

Out[34]:

	exam_score	grade
0	59.197075	C
1	98.718319	A
2	33.361684	F
3	27.830878	F
4	60.686418	C
...	...	...
1148	31.321091	F
1149	45.230354	D
1150	70.239270	B
1151	55.382948	C
1152	67.190582	C

1153 rows × 2 columns

In [35]:
```
# question 15
df['age_group'] = pd.cut(
    df['age'],
    bins=[15, 18, 21, 24, 35],
    labels=['15-18', '19-21', '22-24', '25+'],
    include_lowest=True,
    right=True
)

df[['age', 'age_group']]
```

Out[35]:

	age	age_group
0	21.048994	22-24
1	20.496321	19-21
2	20.496321	19-21
3	25.350557	25+
4	17.937754	15-18
...	...	...
1148	20.169749	19-21
1149	22.671669	22-24
1150	16.382979	15-18
1151	20.676107	19-21
1152	24.716932	25+

1153 rows × 2 columns